

Ritik Jain

GitHub: github.com/rjain2470

Email: ritik.jain@uconn.edu

LinkedIn: [linkedin.com/in/ritik-jain-91a201220/](https://www.linkedin.com/in/ritik-jain-91a201220/)

EDUCATION

- University of Connecticut** Storrs, Connecticut
Ph.D. in Mathematics; GPA: 3.89 August 2024 – Present
- Fordham University** New York, New York
B.S. in Applied Mathematics, summa cum laude; GPA: 3.97 August 2021 – May 2024

PUBLICATIONS

- R. Jain**, H.-B. Moon, and P. Wu, “Distribution of the number of zeros of polynomials over a finite field,” *Involve* 18-4 707-718 (2025). Preprint: [arXiv:2308.14580](https://arxiv.org/abs/2308.14580). DOI: doi.org/10.2140/involve.2025.18.707
- R. Jain**, “The number of solutions of a random system of polynomials over a finite field,” (2024) Preprint: [arxiv:2409.06866](https://arxiv.org/abs/2409.06866).

TEACHING

University of Connecticut

- Calculus I, Teaching Assistant Spring 2026
- Calculus I, Teaching Assistant Fall 2025
- Math for Business and Economics, Teaching Assistant Summer 2025
- Calculus II, Teaching Assistant Spring 2025
- Math for Business and Economics, Teaching Assistant Fall 2024

PROJECTS

- Neumann-prover**: Developed a state-of-the-art autoformalization pipeline powered by 20+ foundation models (OpenAI, Anthropic, etc.). Leveraged dynamic model orchestration and compiler-in-the-loop structured feedback mechanisms to successfully formalize complex theorems, advancing automated reasoning and autoregressive theorem proving.
- RAG-prover**: Embedded mathlib4, a library of over 100k formalized theorems, into Euclidean space using OpenAI’s text-embedding-3-large. Leveraged recent advancements in ML, including FAISS (Facebook AI Similarity Search), for ultra-fast, high-dimensional retrieval. Seamlessly integrated embedding-based retrieval into a Retrieval-Augmented Generation (RAG) pipeline, delivering a significant performance boost in dynamically generating formal proofs in Lean.

PROFESSIONAL EXPERIENCE

- Rogo** New York, NY
Intern August 2022 – May 2023
 - Supported reporting and exploratory analysis with Python; performed market analysis driven by statistical methods and presented research weekly to founders as the sole intern at the company.
 - Prepared investor pitch decks and a five-page strategy memorandum read by founders; cleaned and organized datasets to support model training; produced concise analyses for leadership.
- Steel Key Capital** New York, NY
Intern January 2022 – August 2022
 - Used Python to create an optimized in-house cryptocurrency analysis pipeline using live data, detecting lucrative pairs trading opportunities using correlation matrices.
 - Analyzed financial data for over 10 cryptocurrencies to craft delta-neutral investment strategies for a portfolio valued at over 10 million dollars, guiding investments of valued at \$50,000.
 - Delegated tasks to 6 interns, including investment reports and data analysis.

SKILLS

- Technical Proficiencies**: Python, Github workflows, Jupyter, C++, NumPy, SciPy, PyTorch, pandas.
- General**: Statistics, Mathematics, Machine Learning, Data Science.